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Python Quantum Logic Visualiser Program

Problem and context

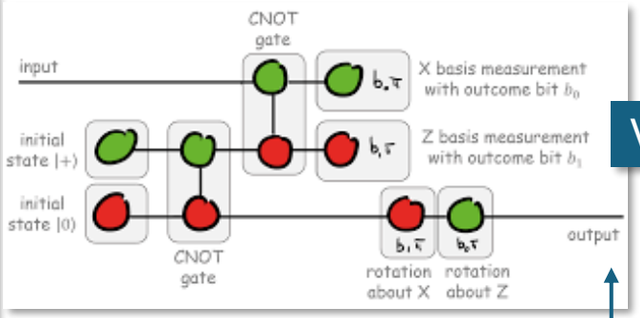
Quantum computing is on the rise, so naturally people want to learn about it and try to understand it. However, most beginner courses jump straight into complicated mathematics that intimidates and deters most avid, but novice learners – especially secondary/high school students.

- When I first tried learning it myself, Bob Coecke's quantum picturalisms made the information significantly easier to digest – especially ZX - style visual reasoning. This fascinated me and made it possible for me to try to understand the basics of quantum logic.
- Therefore, I want to create:

A python tool that turns quantum operations into simple visual diagrams.

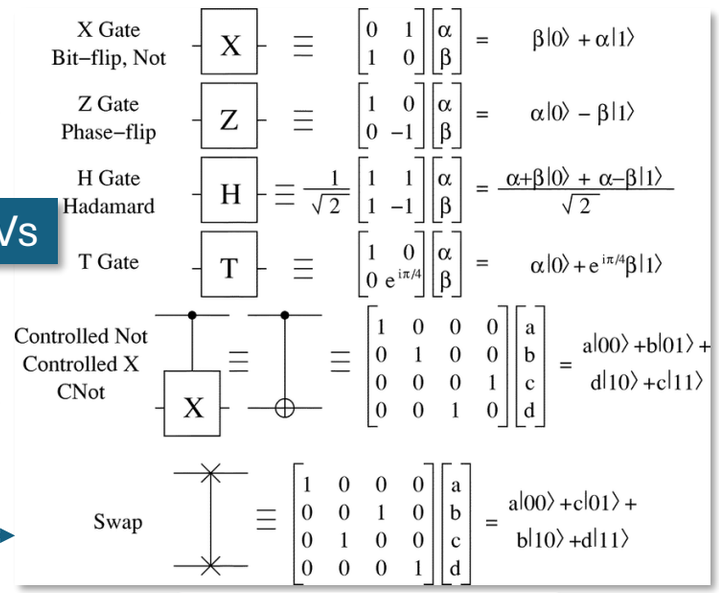
Enabling beginners to see the logic behind quantum computing without having to memorise equations. This tool could also be used by computer science students to visualise quantum gates in a hands-on way, engaging them with intuitive visuals.

- I want to make this python tool based on Bob Coecke's work and integrate quantum picturalism into a python program to make the basics of quantum logic more understandable and easier for high school students and others who want to learn about it.



Quantum Picturalism: Learning Quantum Theory in High School

Vs



Normal quantum gates and their matrix representations

Project plan

Week 1 – Research and sketches

Week 2 – Basic python setup

Week 3 – Gate functions

Week 4 – Diagram engine

Week 5 – Testing

Week 6 – Final prototype

Initial ideas and software components

Core python components:

- Tkinter canvas : to draw circles, lines and animations.
- Gate functions: apply_Hadamard(state)
  apply_x(state)
- State representations: I will store qubit state as simple numbers or labels e.g., superposition, 0/1 prob etc.
- State to visual:
  - Green node for Z
  - Red node for X
  - Split lines for superposition.

User

Gate buttons

Python gate logic

Diagram engine

Tkinter output

Project specification

It must be written entirely in python.

- It must run on any laptop.
- The interface must be made with Tkinter for the basic animation.
- It must support a small set of beginner gates.
- These gates could be:
  - X (NOT)
  - Z
  - H (Hadamard)
  - CNOT
- It must show each gate's effect as a diagram changing shape/colour.
- It must include a step though mode or option to show the transformations slowly.

Challenges

- I have never animated diagrams in python before.
- Translating quantum ideas without maths is difficult.
- Getting timings and animations smooth in Tkinter.

How I'll overcome them

- Start with static diagrams then add animation later if that goes smoothly.
- Test on parents and friends to see which parts aren't clear.
- Keep the animations simple and focus on clarity.

Impact

- Makes quantum computing accessible to young learners.
- Could be used in STEM clubs or online tutorials to increase engagement in STEM globally, making it accessible to all.
- Helps more people understand quantum logic.

Initial research

- I looked at Python graphics libraries. Tkinter is simplest for buttons and shapes so I will start using that and see if it is suitable.
- I also looked at some existing python simulators that showed matrices and they looked very complex.
- I started reading 'Quantum Picturalism' – the paper linked above – however realised I would benefit from a deeper understanding of ZX-style diagrams, so I am currently reading 'Quantum In Pictures'.
- Due to my research, I concluded that there is a gap for a beginner's friendly, visual quantum explainer.

Design brief

- Build a beginner friendly python program that visually shows how quantum gates change a qubit state.
- Possibly allows the user to build small circuits and see the visual transformation step-by-step (animation).
- Uses simple shapes (circles, lines and arrows) – as shown in the picture above – instead of mathematical notation like, for example, Hilbert representations. Focuses on visually explaining; what does this gate *do* and *why?*

Input gate → Diagram → Output state